

Name

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- Write down the gen solution of the ODE $y' = A y$ if A has eigenvalues $\lambda_1 = -1, \lambda_2 = -3, \lambda_3 = -4$ with matching eigenvectors v_i . Which part of the solution matters most as $t \rightarrow \infty$?

 $y(t) =$

22

1. Write down the gen solution of the ODE $y' = Ay$ where $A = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ and give the most important part as $t \rightarrow \infty$

 $y(t) =$

22

- Write down the gen solution of ODE $y' = A y$ where

$$A = \begin{pmatrix} -1 & 2 & 0 \\ 2 & -4 & 0 \\ 0 & 0 & 0.1 \end{pmatrix}$$
 and give the most important part as $t \rightarrow \infty$.

$y(t) =$	z	
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- 2.** Write down the gen solution of ODE $y' = A y$ where

$$A = \begin{pmatrix} -2 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$
 and give the most important part as $t \rightarrow \infty$.

$y(t) =$	$\approx$	
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